

Sales and Profitability Review

Power BI project | Microsoft Financial Sample

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I developed this project to identify products and markets with substantial sales but weaker profit margins. The full sample highlights Velo and the United States as areas for review, while the Enterprise segment records a loss. These findings help prioritize questions about costs, pricing and discount practices.

Business question and scope

Which products and markets generate strong sales but weak profit margins?

Where should a stakeholder investigate costs or discounts?

The intended reader is a sales or finance stakeholder who needs a clear overview and the ability to explore individual countries, products and reporting periods.

The dataset contains 700 records, 6 products, 5 countries and 5 segments across 16 months from September 2013 through December 2014. Findings below include all records, with country and year filters cleared and no selected chart points. Monetary values use the sample's dollar notation (not identified as CAD).

Preparation in Power Query

I imported the Financials table and prepared it in Power Query before creating the report. I kept all products and negative profits so that weak performance and losses remained visible, and I retained repeated-looking records without assuming they were duplicates.

- Set categorical fields to Text and trimmed unnecessary spaces in Country, Product, Segment and Discount Band.

- Set Date to Date, financial amounts to Fixed Decimal Number, and Month Number and Year to Whole Number. Kept Units Sold as Decimal Number to preserve fractional quantities.

- Inspected column quality for missing values and errors, particularly in Date, Product, Country, Sales, COGS and Profit.

- Added Month Start using `Date.StartOfMonth([Date])` so that reporting groups preserve both month and year.

- Added Profit Difference using `([Sales] - [COGS]) - [Profit]`. Corrected a field-reference error by inserting the actual column names from Available columns, then standardized readable headers.

Applied the transformations and saved the Power BI project as Sales-Profitability-Review.pbix.

These steps document the preparation process and can run again when the source is refreshed. COGS means cost of goods sold, the direct cost associated with the products sold.

Measures and overall results

The six DAX measures calculate totals and ratios within the selected filter context. Profit Margin is total profit divided by total sales, and Discount Rate is total discounts divided by total gross sales. Using ratios of totals avoids giving small and large records equal weight.

KPI	Full sample result
Total Sales	\$118,726,350.26
Total Profit	\$16,893,702.26
Profit Margin	14.2%
Discount Rate	7.2%

DAX definitions

Total Sales = SUM(Financials[Sales])

Total Profit = SUM(Financials[Profit])

Total Gross Sales = SUM(Financials[Gross Sales])

Total Discounts = SUM(Financials[Discounts])

Profit Margin = DIVIDE([Total Profit], [Total Sales])

Discount Rate = DIVIDE([Total Discounts], [Total Gross Sales])

Report design

The report uses four KPI cards for the overall position and a product matrix with Total Sales, Total Profit, Profit Margin and Discount Rate. The Country slicer supports market comparisons without requiring separate reports for each country.

The monthly trend uses Month Start on the X-axis and Total Profit on the Y-axis, with sales, margin and discount rate in the tooltips. The segment comparison places Profit Margin and Discount Rate together in a clustered column chart. A Year slicer supports period selection while preserving the month and year in the trend.

Reconciliation and final checks

An independent calculation from the original sample confirms Sales - COGS - Profit equals zero for every record. Gross Sales - Discounts - Sales also equals zero for every record. The source totals are \$127,931,598.50 gross sales, \$9,205,248.24 discounts and \$101,832,648.00 COGS.

The source contains no blank cells. The Discount Band value “None” is a category indicating no discount, rather than a missing value. Keeping decimal quantities also preserves the 36 records with fractional Units Sold.

Before publishing, compare the four cards with the full sample results above, test Country and Year selections, and confirm that each visual updates. Clear filters and chart selections before taking the dashboard screenshot, then save the final file.

Findings and recommended follow up

Velo combines substantial sales with the weakest product margin

Velo ranks third in sales at \$18.25 million, but its 12.6% profit margin is the lowest of the six products. It’s also below the overall 14.2%. It also has the highest product discount rate at 8.0%. I would investigate Velo’s pricing, cost of goods sold and discount patterns by country and segment before recommending a change. Paseo leads sales at \$33.01 million with a 14.5% margin, providing a useful comparison.

Small Business merits review and Enterprise requires cost analysis

Small Business generates \$42.43 million in sales with a 9.8% margin and a 7.6% discount rate. Its margin is below the overall rate, while its discount rate is higher. However, Midmarket has a higher discount rate of 7.8% and a much stronger 27.7% margin. Discounts alone therefore do not explain the segment differences.

Enterprise generates \$19.61 million in sales but loses approximately \$614,546, giving a -3.1% margin. Its discount rate is 6.9%, below the overall rate. COGS of \$20.23 million exceeds sales, so a stakeholder should examine the segment’s cost structure, prices and product mix. Retaining losses in the dataset makes this issue visible.

The United States has the highest sales and lowest country margin

The United States generates \$25.03 million in sales, the highest country total, but has the lowest country margin at 12.0% and the highest country discount rate at 8.2%. Germany achieves a 15.7% margin and a 5.7% discount rate. Comparing product and segment mix across these markets would help assess whether costs, prices or discount practices warrant attention.

Monthly profit varies and the strongest month is not the sales peak

December 2014 has the highest profit at \$2.03 million and a 16.9% margin. November 2014 has the lowest profit at approximately \$604,600 and an 11.2% margin. October 2014 has the highest sales at \$12.38 million, showing that the sales peak and profit peak occur in different months. These observations support a review of monthly costs and sales mix.

Two sentence stakeholder summary

Velo generated \$18.25 million in sales with the lowest product margin of 12.6%, while the Enterprise segment lost approximately \$614,546 despite \$19.61 million in sales. A stakeholder

should prioritize reviewing Velo's pricing and discounts and Enterprise's cost of goods sold and product mix, using country and year filters to narrow the investigation.

Limitations and source

This project uses Microsoft sample data and does not describe Tourmaline's operations. Profit is Sales minus COGS in this dataset, so it does not represent net profit after other operating expenses. Only September through December is available for 2013, so full-year totals are not directly comparable with 2014, and the data is insufficient to establish recurring seasonality.

Source: Microsoft Financial Sample Excel workbook. Calculations use all 700 original records.
<https://learn.microsoft.com/en-us/power-bi/create-reports/sample-financial-downlo>